

# Correction to a Typo on Page 147 of Fourier Analysis by Stein and Shakarchi

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July 2026

## 1 Footnote

*The author is a high school student from Singapore who discovered this typo while reading the book. Readers are therefore advised to verify all statements made in this text.*

This entire article is about a minor typographical error on page 147 of Stein and Shakarchi's *Fourier Analysis*. Specifically, in the proof of Theorem 2.1 (iii) in Chapter 5, the following equation, due to the Plancherel Theorem, is written:

$$\begin{aligned}\int_{-\infty}^{\infty} |u(x, t) - f(x)|^2 dx &= \int_{-\infty}^{\infty} |\hat{u}(\xi, t) - \hat{f}(\xi)|^2 d\xi \\ &= \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1| d\xi.\end{aligned}$$

The second line above is incorrect because

$$|e^{-4\pi^2 t \xi^2} - 1| \neq |e^{-4\pi^2 t \xi^2} - 1|^2.$$

Fortunately, this does not affect the line of argument in the proof, as the decay condition on the imposed Fourier transform ensures that their estimates hold.

The updated proof is written below for completeness, though the same method is used.

## 2 Updated Proof:

The Plancherel Theorem yields

$$\begin{aligned} \int_{-\infty}^{\infty} |u(x, t) - f(x)|^2 dx &= \int_{-\infty}^{\infty} |\hat{u}(\xi, t) - \hat{f}(\xi)|^2 d\xi \\ &= \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1|^2 d\xi. \end{aligned}$$

We want to show that the last integral goes to zero as  $t \rightarrow 0$ . Note the bound  $|e^{-4\pi^2 t \xi^2} - 1| \leq 2$ . As  $f \in S(\mathbb{R})$ , for every  $\varepsilon > 0$ , we may find  $N \in \mathbb{N}$  sufficiently large such that

$$\int_{|\xi| \geq N} |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1|^2 d\xi < \frac{\varepsilon}{2}.$$

Moreover, we may choose  $t$  sufficiently small so that

$$\sup_{|\xi| \leq N} |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1|^2 < \frac{\varepsilon}{4N}$$

as  $\hat{f}$  is bounded.

Thus, we have the estimate

$$\begin{aligned} \int_{|\xi| \leq N} |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1|^2 d\xi &\leq \int_{|\xi| \leq N} \left( \sup_{|\xi| \leq N} |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1|^2 \right) d\xi \\ &< \frac{\varepsilon}{4N} (2N) = \frac{\varepsilon}{2}. \end{aligned}$$

for all sufficiently small  $t$ , which completes the proof of the theorem. ■